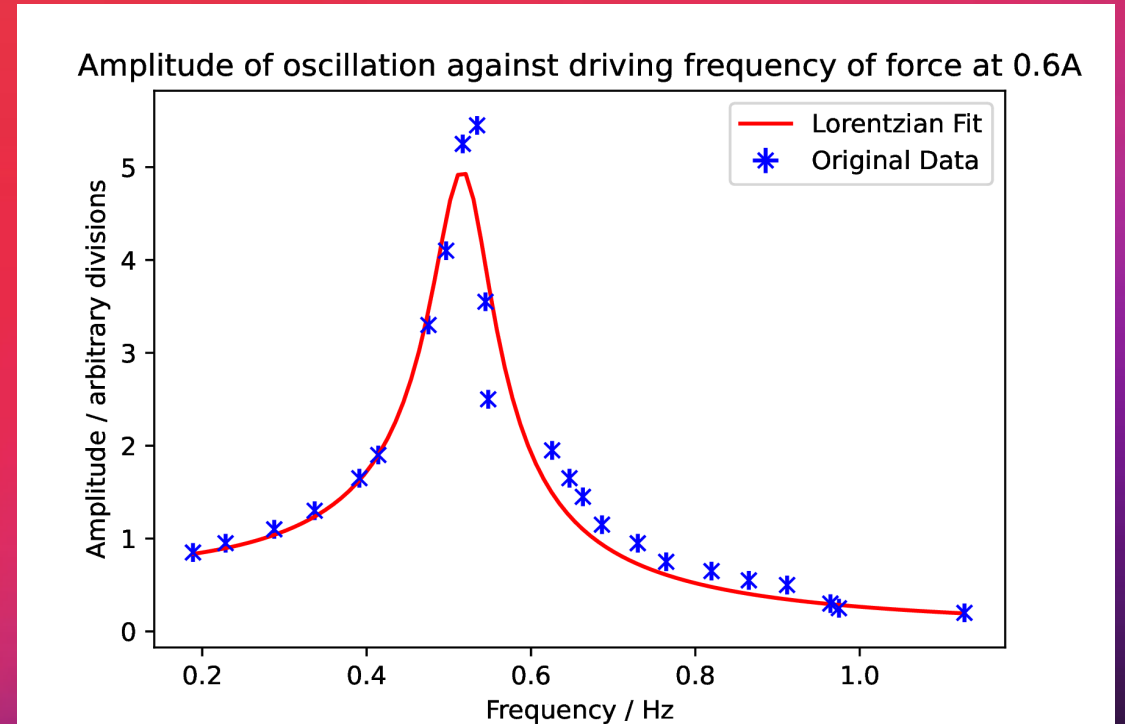


Resonance

From the resonance peak, the resonant angular driving frequency, ω_D , is $(3.26 \pm 0.02) \text{ rad s}^{-1}$.

Using the resonance shift equation (5), we get a value for the natural frequency of $(3.28 \pm 0.01) \text{ rad s}^{-1}$. This is lower than the predicted value from free oscillations $[(3.33 \pm 0.01) \text{ rad s}^{-1}]$.



A resonance curve with Lorentzian fit for a damping current of 0.6 A. The peak falls at $(0.519 \pm 0.004) \text{ Hz}$.